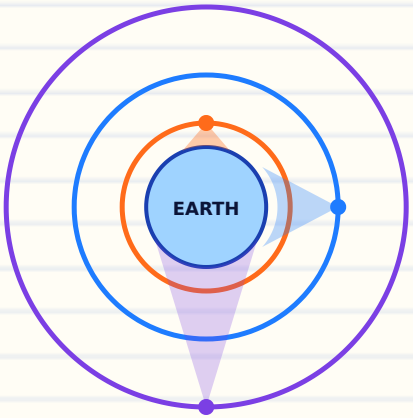
SEC 3 · LEC 2 · EARTH ORBITS + CONSTELLATIONS

Higher = sees more, moves slower, sees less detail

Left = your left notebook page. Right = your right page. You already have LEO/MEO/GEO altitudes from Section 1; the new part is periods, the trade-off, and why constellations exist.

LEFT PAGE · DRAW

S3 / L2 — Earth orbits **date · pg __**



LEO 200–2,000 km
period 1.5–2 h · ISS 400 km, 92 min
tiny footprint, sharp detail, fast

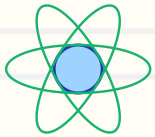
MEO 2,000–35,786 km
period 2–24 h · GPS 20,200 km, 12 h
navigation + comms

GEO 35,786 km
period 24 h = Earth's spin
hangs over one spot · over EQUATOR
sees 1/3 of Earth at once

THE TRADE-OFF (higher ↑)
↑ coverage (bigger footprint)
↓ speed · ↓ detail · ↑ radio delay

LEO sees one spot for minutes → need MANY sats = a CONSTELLATION (Iridium 66 · GPS 24+ · Starlink 1000s)

GEO: 3 sats ≈ whole Earth (not the poles)



constellation: many rings, someone always overhead

⭐ **3 TAKEAWAYS** 3 lines

1. Higher orbit = sees more, moves slower, sees less detail, longer radio delay
2. 35,786 km is the one altitude where one lap = one Earth spin, so GEO hangs still
3. LEO can't watch one place for long, so you fly a swarm: a constellation

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions) **NOTES (keywords)**

3 Earth-orbit bands: altitude + period of each?

Why exactly 35,786 km and not a round number?

Go higher: what gets better, what gets worse?

Why does LEO need a constellation and GEO doesn't?

ISS period? GPS period?

GEO has to sit over what? Why?

Name two constellations and what they're for.

? how many GEO sats to cover the whole planet

Earth orbit = goes around Earth, not the Sun (like the Moon does)
LEO 200–2,000 km · 1.5–2 h · ISS 400 km / 92 min · close, fast, small view, sharp detail
MEO 2,000–35,786 km · 2–24 h · GPS 20,200 km / 12 h · navigation + comms
GEO 35,786 km · 24 h (really 23 h 56 m) · matches Earth's spin → hangs over one spot · must be over the equator
why 35,786: the one height where one lap = one Earth turn (higher = slower, Kepler 3 later)
trade-off: ↑ altitude → bigger view, slower, blurrier, more radio delay · ↓ altitude → sharp, fast, tiny view, drag
LEO sees one spot for minutes → fly many = constellation (Iridium 66 @ 780 km · Starlink 1000s @ 550)
GPS: 24+ sats, 6 rings, 12 h → ≥ 4 always overhead → your phone finds itself · Russia GLONASS, EU Galileo, China BeiDou
3 GEO sats ≈ whole Earth (poles excluded)
no "best" orbit: pick by mission (detail vs coverage vs delay vs cost) · instructor's "36,000 km from center" is wrong: 35,786 is ALTITUDE; from center ≈ 42,164

SEC 3 · LEC 2 · TEACH + FACT CHECK

Three cameras over a football field

Start with the box. Then the detail, the caption errors, and where the instructor was loose.

🧒 Like you're 5

Imagine filming a football game three ways. A **drone ten feet up** sees one player in crisp detail, but it has to zip around fast and it only ever sees one little patch. That's LEO. A camera on a **tall pole** sees half the field, a bit blurrier, and moves slower. That's MEO. A **blimp so high** it sees the whole stadium at once, tiny and blurry, and it just hangs there. That's GEO.

Now: what if you want crisp detail *and* the whole field *all the time*? One drone can't do it. So you fly a **swarm of drones** taking turns, so there's always one over every part of the field. That swarm is a **constellation**. GPS is a swarm. Starlink is a swarm. Iridium is a swarm.

🔭 The three bands, and the one rule behind them

- **The rule:** the higher you orbit, the slower you go and the longer one lap takes. LEO laps in 90 minutes. GPS laps in 12 hours. GEO laps in exactly one day. (Kepler's third law, which the next lecture will name.)
- **LEO, 200–2,000 km, 1.5–2 h.** Close means sharp pictures and short radio hops, but a tiny view and only a few minutes over any one place. The ISS at 400 km passes over Florida and is back 92 minutes later. Most satellites ever launched live here.
- **MEO, 2,000–35,786 km, 2–24 h.** The middle ground. GPS sits at 20,200 km with a 12-hour lap: high enough that a dozen satellites see most of the planet, low enough that the signal isn't too weak or too late. Russia (GLONASS), Europe (Galileo), and China (BeiDou) run their own.
- **GEO, 35,786 km, 24 h.** One lap takes exactly one Earth rotation, so from the ground the satellite never moves. TV dishes bolt to a wall and never adjust. It must sit over the equator; anywhere else, the ground would slide underneath it as Earth turns. One GEO satellite sees about a third of the planet; three cover nearly all of it except the poles.
- **Why 35,786 exactly?** It's not chosen, it's calculated. It's the only altitude where the lap time equals one Earth spin. Lower is too fast, higher is too slow. Callback to your Section 1 page: you wrote this number down before you knew why.

🛰️ Constellations

- **Iridium:** 66 satellites at 780 km in 6 rings over the poles. Satellite phones that work anywhere on Earth, including the oceans.
- **GPS:** 24 required, ~31 flying, 6 rings tilted 55°. Arranged so at least 4 are above you at all times, which is the minimum your phone needs to compute a position.
- **Starlink:** thousands at ~550 km for internet. Same idea at a much bigger scale.

🔎 Caption errors

Captions said	Instructor meant
"meal"	**MEO**
"The Sims or your entity"	"it **seems arbitrary**" (200, 2,000 look round; why 35,786?)
"Leo's"	LEOs
"the audio of the GEO orbit"	the **idea** of the GEO orbit
"lower level bit"	low Earth orbit
"care a compromise"	**make** a compromise

▲ Where the instructor was loose

- **"Thirty-four, thirty-six thousand km from the center of Earth."** No. 35,786 km is the *altitude* above the surface. From the center it's about 42,164 km (add Earth's 6,378 km radius). He mixed the two, which is exactly the mistake your Section 1 notes warned about.
- **"GEO period 24 hours."** Close enough for conversation; the exact value is 23 h 56 min, one true rotation relative to the stars (the sidereal day from your Section 1 vocab). Four minutes matters to the people who park the satellite.
- **"At 1,000–2,000 km, 5, 6, 10-hour period."** Muddled. At 2,000 km the lap is about 2 hours. The 5–10-hour laps are up around 8,000–15,000 km. The slide's "2 to 24 h" range is right; the spoken example wasn't.
- **"Always remain at the same altitude, so the same relative position."** Constant altitude is true of any circular orbit. What makes GEO special is constant position *over the ground*, which comes from matching Earth's spin, not from altitude staying fixed.
- **Correct and worth keeping:** the slide picture isn't to scale, and

SEC 3 · LEC 2 · VOCABULARY

Every word used, in plain English

Several of these are repeats from Section 1. That's deliberate: say them again out loud. Copy only the new ones into the word bank.

Earth orbit — any path that circles Earth rather than the Sun. The Moon is in one; so is every satellite. *Opposite of an interplanetary trajectory.*
Period — the time for one complete lap. 92 minutes for the ISS, 12 hours for GPS, one day for GEO. *The orbit's "day."*
LEO / MEO / GEO — low, medium, geostationary Earth orbit: the three altitude bands (200–2,000 km, 2,000–35,786 km, 35,786 km). *Downstairs, middle floor, penthouse.*
Geostationary — a GEO orbit that is also over the equator, so the satellite seems frozen in the sky. *Repeat from Section 1: geosynchronous = one-day lap; geostationary = one-day lap AND over the equator.*
Equatorial plane — the flat sheet through Earth's middle. GEO satellites must lie in it. *The table the hula hoop sits flat on.*
Rotation period / "Earth rotation velocity" — how long Earth takes to spin once: 23 h 56 min against the stars, 24 h against the Sun. A GEO satellite's lap matches it.
Altitude vs distance from center — altitude is height above the ground; distance from center adds Earth's radius (6,378 km). GEO: 35,786 km altitude, 42,164 km from center. *The instructor mixed these up.*
Coverage / footprint — the patch of Earth a satellite can see or talk to at one moment. Tiny for LEO, a third of the planet for GEO. *The circle of light from a flashlight: closer = smaller circle.*
Global coverage — being able to see or serve every point on Earth, all the time. One satellite can't; a constellation can.
Resolution — how much detail a camera can make out. Closer = better. A LEO satellite can see a car; a GEO weather satellite sees whole storm systems. *Reading a sign from 10 feet vs 10 miles.*
Constellation — a group of satellites in planned orbits that work as one system, so at least one is always where it's needed. *A swarm of drones taking turns.*
Orbital plane / "ring" — the flat sheet one orbit lies in. Constellations spread satellites across several planes (GPS: 6) so they're not all bunched on one side of Earth.
Navigation satellite — broadcasts a precise time signal; your receiver hears several and triangulates where it is. Needs at least 4 overhead. *Repeat from Section 1.*
GPS / GLONASS / Galileo / BeiDou — the American, Russian, European, and Chinese navigation constellations. All in MEO around 20,000 km. Your phone uses several at once.
Iridium — a company and its constellation: 66 satellites at 780 km giving satellite-phone coverage everywhere on Earth. Named after element 77 because the original plan was 77 satellites.
Starlink — SpaceX's internet constellation, thousands of satellites at ~550 km. *Repeat from Section 1.*
ISS — International Space Station, 400 km, 92-minute lap. *Repeat.*
Data relay / transfer time — passing information from satellite to ground (or satellite to satellite). Higher orbit = longer radio trip = more delay. GEO adds about a quarter second each way.
Trade-off / compromise — you can't have every good thing at once; improving one (coverage) costs another (detail, delay). Orbit choice is a trade-off. *Big truck vs sports car.*
Optimize — tune the design to get the best balance for what the mission actually needs. There's no universally "best" orbit, only the best for this job.
Give or take — spoken shorthand for "approximately." He said LEO periods are "1.5 to 2 hours, give or take."

SEC 3 · LEC 2 · SAY IT OUT LOUD

Page 4: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I fill the box.

🗣 The friend's opening question

"Why are some satellites really low and some really high? Does it matter?"

✅ Three sentences you should be able to say

- The higher you go, the more of Earth you can see at once, but you move slower, you see less detail, and your radio signal takes longer.
- There's one height, about 36,000 km, where one lap takes exactly one day, so the satellite just hangs over the same spot. That's where TV and weather satellites sit, always over the equator.
- Low satellites only see a tiny patch for a few minutes, so if you want to cover the whole planet you fly a whole swarm of them. That's what GPS and Starlink are.

❓ Follow-ups a friend might ask

- "So why doesn't the ISS just hang over one spot too?" (it's too low; one lap is 92 min, so it's gone in minutes)
- "Why does the GPS one have to be in the middle?" (high enough that a few dozen see the whole planet, low enough the signal's strong and fresh)
- "How many satellites does my phone need to see?" (at least 4)
- "Why over the equator? Couldn't it hang over New York?" (Earth's spin would drag New York away underneath; only the equator moves in a matching circle)
- "Is 36,000 km from the ground or from the middle of the Earth?" (ground; from the middle add ~6,400)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Review link: this lecture repeats your Section 1 satellite page. Flip back to it first and see what you can still answer cold.